

Albany County Behavioral EV Charging Results

Full dataset run with ZIP/ZCTA adoption scaling and H3-8 charging geography

NYSERDA EV Adoption and Charging Model

June 16, 2026

This week's status

Area	Current status
Model geography	Adoption now scales from home ZIP/ZCTA; charging load is placed at destination H3.
Charging behavior	Allocation now uses conditional expected value instead of simple expected-capacity scaling.
Managed charging	Applied only to charger types marked as managed-eligible in assumptions.

Remaining trip-chain modeling decision: final stops

Final activity distribution for chains that do not end at home

Final activity type	Chains
Social / Recreational	7,572
Work	3,300
School / Daycare / Religious	1,620
Shopping / Errands	909
Other	715
Transport / drop-off / pick-up	450
Meals	117
Medical / Dental	105

- ▶ **As observed:** final non-home stops remain non-home overnight stops.
- ▶ **Completed chain:** add a synthetic return-home leg and overnight home dwell.

Geography design: home adoption vs. physical load

Core decision

Keep home adoption geography separate from charging-load geography.

Use case	Target geography	Reason
EV adoption forecast	Home ZIP/ZCTA	Adoption model is ZIP-level; join from home coordinate to ZIP/ZCTA.
Physical charging load	Destination H3	Load appears where vehicles charge, not necessarily where they live.

Charging allocation update

Old risk

Expected capacity logic could under- or over-allocate energy when a lower-priority stop had partial capacity and uncertain availability.

Current method

Carry probability states across ranked charging opportunities:

$$\mathcal{S}_j = \{(p_k, r_k) : \text{probability } p_k, \text{ remaining energy } r_k\}$$

At each stop, split every state into available and unavailable branches, allocate:

$$E_{j,k} = \min(r_k, C_j)$$

and aggregate expected event energy:

$$\mathbb{E}[E_j] = \sum_k p_k \cdot q_j \cdot E_{j,k}$$

where C_j is stop capacity and q_j is the GridUp-style charger availability:

$$q_j = \min \left(1, q_j^{\text{peak}} \cdot \frac{\text{peak stopped vehicles}}{\text{current stopped vehicles}} \right).$$

Managed vs. unmanaged charging

Unmanaged

Charging begins at arrival and uses charger power until allocated energy is delivered or the dwell window closes:

$$T_s = \min \left(\tau_s, \frac{E_s}{P_s} \right)$$

Managed

Only managed-eligible charger types are spread across the dwell window:

$$\bar{P}_s = \min \left(P_s, \frac{E_s}{\tau_s} \right)$$

- ▶ Managed charging applies to:
 - ▶ home L2: 7.2 kW, managed eligible;
 - ▶ work L2: 7.2 kW, managed eligible
 - ▶ civic, long public, and quick public: not managed eligible
- ▶ Availability is now GridUp-style: SFH home L2 has 100% peak-hour probability, MFH home L2 and work L2 use 50%, then probabilities rise in lower-demand hours.
- ▶ This keeps home/work management separate from public/DCFC charging behavior.

Adoption scaling logic

Forecast scaling equation

$$L_{g,t,c,y} = \sum_h L_{h,g,t,c}^{\text{fullEV}} \cdot A_{h,y,s} \cdot G_{h,y,s}$$

Symbol	Meaning
h	Home ZIP/ZCTA used for adoption attribution.
g	Charging H3 cell where load physically appears.
c	Charger category / charging location type.
$A_{h,y,s}$	Adoption fraction for home geography, year, and scenario.
$G_{h,y,s}$	Vehicle growth factor for home geography, year, and scenario.

Result Summary

Question	Current result
Full-EV daily energy delivered	3.305 GWh per modeled day under both managed and unmanaged modes.
Full-EV peak load	287.7 MW unmanaged at 20:00; 197.8 MW managed at 03:00.
2027 baseline adoption load	155.0 MWh per modeled day; 13.49 MW unmanaged peak.
Managed charging effect	31.3% lower full-EV peak and 31.3% lower 2027 baseline peak.

Takeaway: The model now produces a home-ZCTA by charging-H3 by time load surface, then scales it with the ZIP-level adoption forecast.

Modeling Geography

Adoption scaling geography

- ▶ Adoption forecast is ZIP/ZCTA-level.
- ▶ Each modeled vehicle day keeps its home ZCTA.
- ▶ Scaling is applied by `home_zcta`, forecast year, and adoption scenario.

Charging-load geography

- ▶ Charging load is placed at the destination charging location.
- ▶ Primary planning geography is H3 at resolution 8.

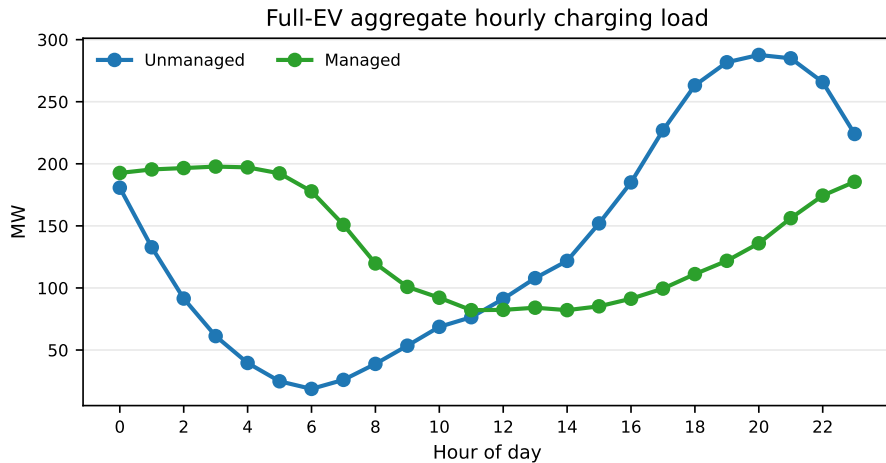
Full-EV Energy Accounting

Metric	Value per modeled day
Vehicle days	216,960
Total network miles	8.964 million miles
Mean daily network miles	41.3 miles per vehicle day
Daily travel energy need	3.765 GWh
Expected charging energy delivered	3.305 GWh
Unmet energy	0.460 GWh
Unmet share of daily travel energy	12.2%

Interpretation

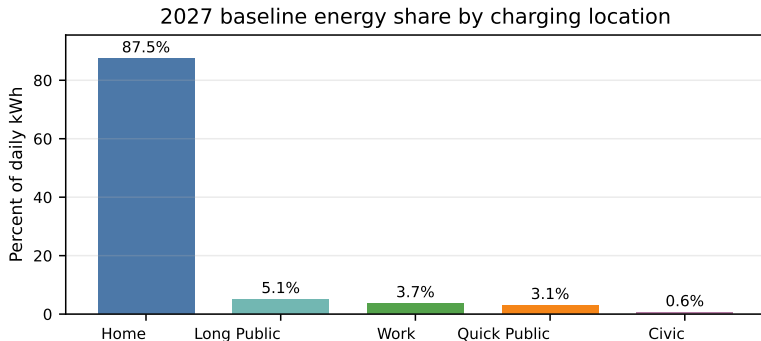
The unmet-energy metric is not a grid load. It is a behavioral feasibility signal showing where the current stop, dwell, charger-access, and power assumptions cannot fully replenish modeled daily travel energy.

Full-EV Hourly Load Shape



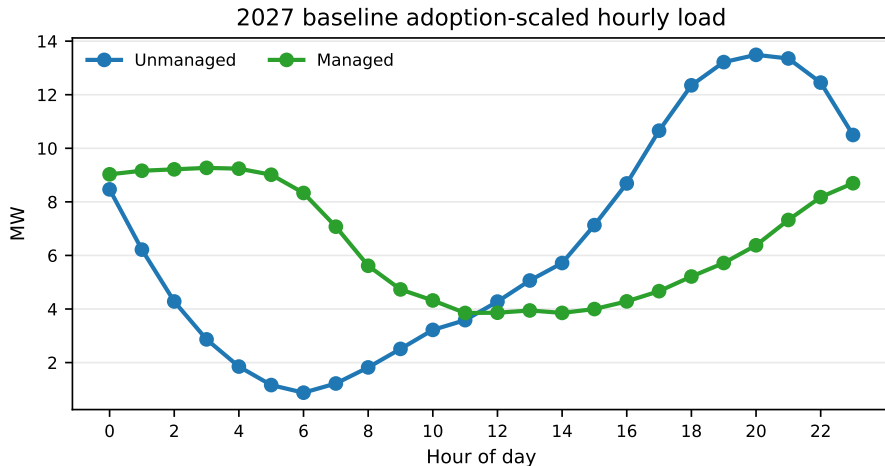
Takeaway: Managed charging shifts the aggregate peak from 20:00 to 03:00 and reduces the full-EV peak from 287.7 MW to 197.8 MW.

Charging Energy Is Concentrated at Home



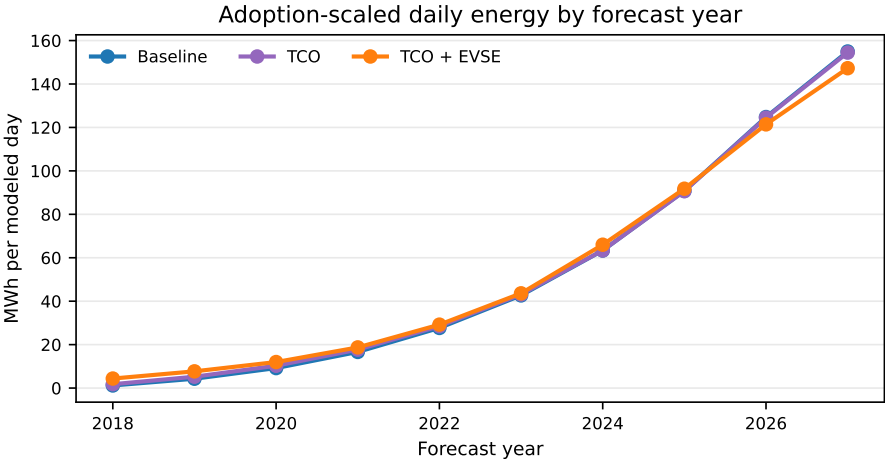
Takeaway: Home charging accounts for 87.5% of 2027 baseline daily energy; public, work, and civic charging together account for the remaining 12.5%.

2027 Baseline Adoption-Scaled Load

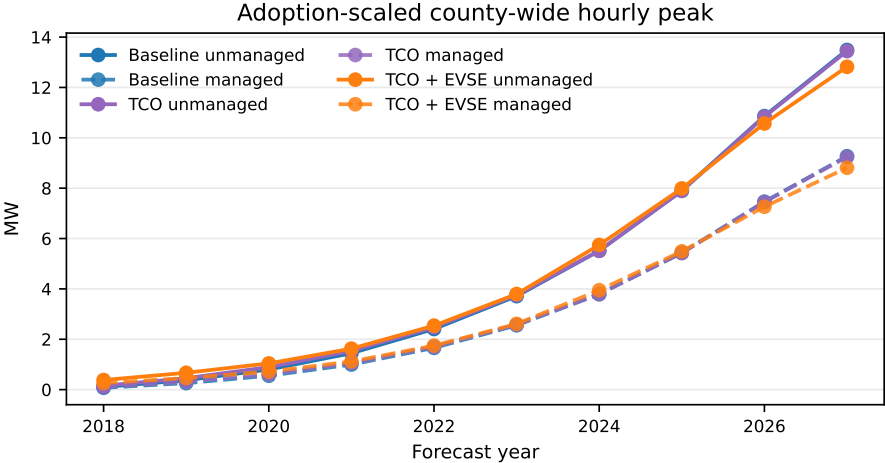


Takeaway: For 2027 baseline adoption, managed charging lowers the county-wide hourly peak from 13.49 MW to 9.27 MW, with the same total daily energy of 155.0 MWh.

Adoption-Scaled Energy by Forecast Year



Adoption-Scaled Peak by Forecast Year



2027 Baseline H3 Hotspots

Unmanaged top H3-hours

H3 cell	Hour	kW
882b8902b1fffff	20	162.7
882b8902b1fffff	19	157.4
882b890297fffff	20	155.2
882b890025fffff	19	154.2
882b8902b1fffff	21	153.3

Managed top H3-hours

H3 cell	Hour	kW
882b8902b1fffff	4	108.7
882b8902b1fffff	3	108.7
882b8902b1fffff	2	107.0
882b890297fffff	3	106.5
882b8902b1fffff	1	106.4

Current Caveats

- ▶ Charger-access and charger-power assumptions are configurable but not yet calibrated against local charger inventory and housing data.
- ▶ The adoption forecast in this run covers 2018–2027, so longer-term grid planning requires a longer adoption horizon.

Next Steps

1. Map charging H3 cells to feeders, substations, or utility service polygons when those layers become available.
2. Calibrate charger availability by home type, workplace access, and public EVSE inventory.
3. Extend the ZIP/ZCTA adoption forecast beyond 2027 before using the outputs for long-horizon planning.
4. Run sensitivity cases for charger access, managed eligibility, DCFC share, and energy efficiency.